MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 7166

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: TR, frequency range 1,220 to 1,365 MHz, nominal peak power rating 1.5 Mw

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	Ebb	Ii	Alt
Unit:	Mw	---	Vdc	μAdc	ft
Maximum:	2.0	0.001	---	200	10,000
Minimum:	---	---	-750	100	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting: See figure 2 and note 1
Weight: 9 pounds (approx)

TEST CONDITIONS:

Parameter:	Incident power	tp1	tp2	prr	Ii	F
Unit:	Mw	μs	μs	pps	μAdc	MHz
Tolerance:	±5%	±0.15	±0.15	---	---	---
Test condition 1:	0.50	4.0	1.0	250	100	F3
Test condition 2:	2.0	4.0	---	250	100	F3

Frequency		
F	MHz	±
1	1,220	1%
2	1,255	1%
3	1,292	1%
4	1,325	1%
5	1,365	1%

GENERAL:

Preproduction sample approval - Required (see note 2)

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Preproduction sample approval inspection							
1011	Humidity	---	---		---	---	---	---
4021	Degradation due to vibration	---	---	t = 1 hour				
---	Mechanical insertion	1,3	---		---	---	---	---
	Quality conformance inspection, part 1	4	---					
4406	Ignitor voltage drop	---	---	Ii = 100 μAdc	Eid	275	475	Vdc
4452	Spike-leakage energy	---	1		Ws	---	0.25	erg
4452	Flat-leakage power	---	1		pf	---	75	mw
1027	Temperature cycling (nonoperating)	---	---		---	---	---	---
4401	Ignitor ignition time	---	---	Ebb = -750 Vdc; Ri = 4.5 Meg	t	---	5.0	sec
4473	Low-level VSWR	5	---	F = (F1, F2, F3, F4, and F5) ±0.1%	σ	---	1.3	---
4488	Arc loss	---	1		La	---	0.3	dB
	Quality conformance inspection, part 2							
4101	Dielectric material strain	---	---		---	---	---	---
4416	Insertion loss	---	---	F = F3 ±5%	Li	---	0.5	dB
4421	Ignitor interaction	---	---	ΔIi = 100 μAdc	ΔLi	---	0.2	dB
4460	Ignitor noise ratio	---	---	Fo = 1,300 MHz; Ii = 100 μAdc	Nr	---	1.3	---
4471	Recovery time	---	2		t	---	20	μs
	Quality conformance inspection, part 3							
---	Life test	6,7	2	Group D: F = F3 ±3%; R = 2.5 Meg	t	1,000	---	hrs

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3 -Continued							
---	Life-test end points:							
4452	Spike-leakage energy	---	1		Ws	---	0.50	erg
4460	Ignitor noise ratio	---	---	Ii = 100 μAdc	Nr	---	1.5	---
4471	Recovery time	---	2		t	---	25	μs
4406	Ignitor voltage drop	---	---	Ii = 100 μAdc	Eid	---	550	Vdc
4416	Insertion loss	---	---		Li	---	0.7	dB
4488	Arc loss	---	1		La	---	0.3	dB
1027	Temperature cycling life test	---	---	10 cycles	---	---	---	---

NOTES:

1. A suitable gasket shall be bolted between the tube and the mounting seat. Tubes shall be supplied with gaskets at both ends.

2. Preproduction sample approval requirements hereby replace any qualification requirements referable to the product covered herein. The term "First article testing" shall be considered as synonomous with the term "Preproduction sample approval testing." All tests applicable herein (including all preproduction sample approval and quality conformance inspection, parts 1, 2, and 3) shall be performed during preproduction sample inspection. A failure of any one tube in any of the tests shall be cause for decision of preproduction sample disapproval.

3. The tube shall be capable of being inserted and removed from its seat a minimum of 15 times with no deterioration in the tube's electrical characteristics.

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

5. A swept frequency method may be used for this test.

6. The TR tube will be terminated with a waveguide to type N transition (VSWR not to exceed 1.1 at test frequency) and a coaxial type N crystal mount connected to the transition. The mount will contain a type 1N21WE crystal which will be terminated by a dc ground return of 100 ±10 ohms. The crystal mount will have a VSWR not to exceed 1.5 when tested with a microwave source at the test frequency which provides a crystal dc current between 0.3 ma to 1.5 ma in a 100 ohms termination. At the beginning of life test, the crystal noise figure (NF receiver) shall not exceed 7.0 dB under the standard 1N21WE test conditions. The actual crystal noise figure at the beginning of life shall be recorded.

7. The crystal noise figure degradation from the initial value shall not exceed the specified limit. One crystal failure will be allowed during the life test. If a crystal failure occurs, a new crystal shall be inserted in the mount and the life test shall continue for a minimum of 100 hours.

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NOTES: -Continued

8. Changes from previous issue. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI
Air Force - 11, 17
DSA - ES

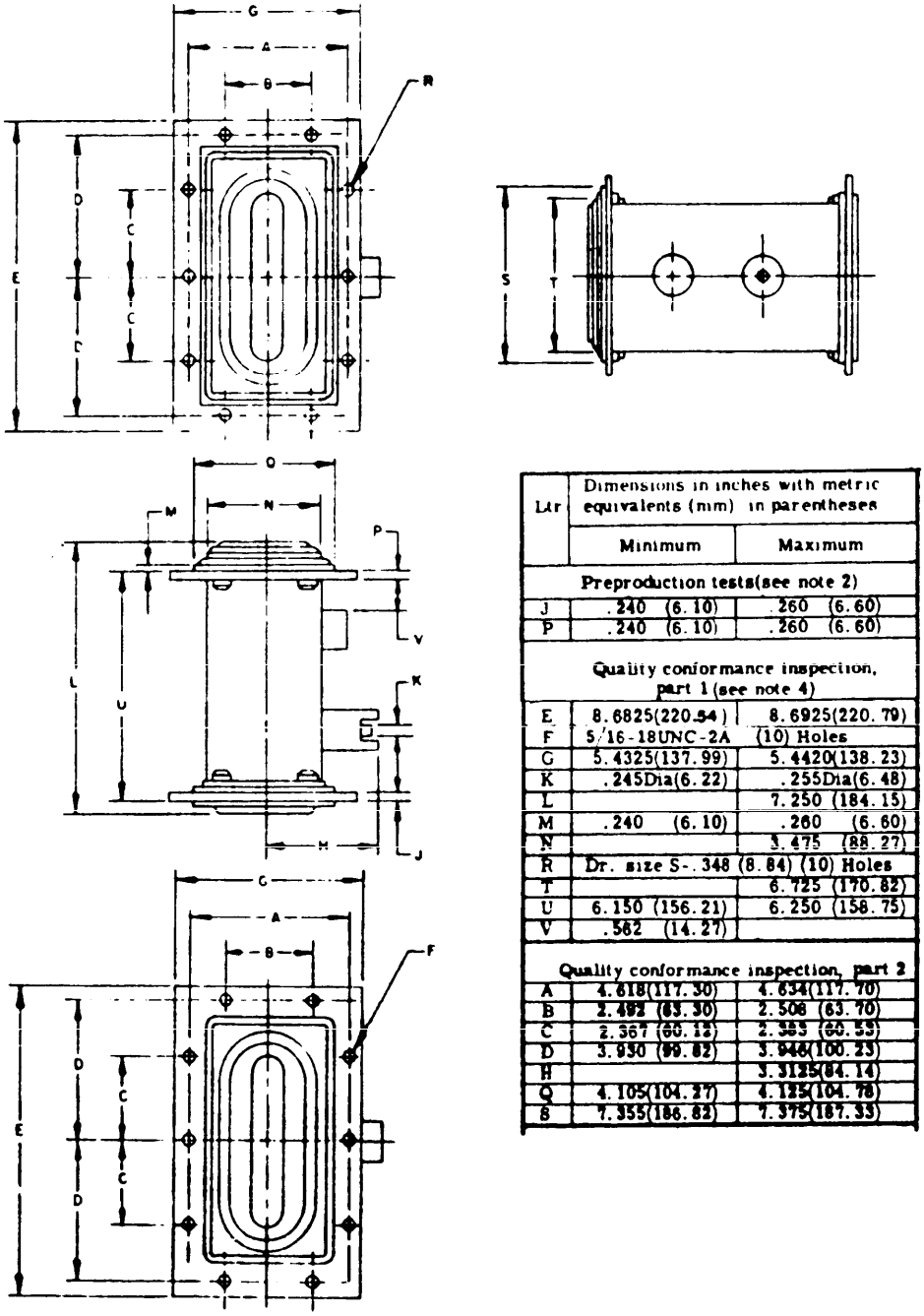
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2617)

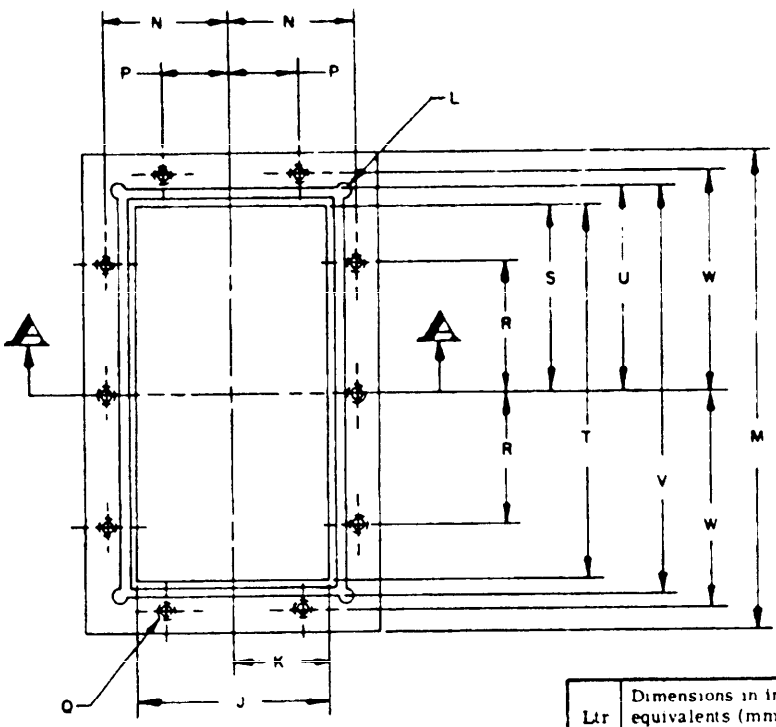
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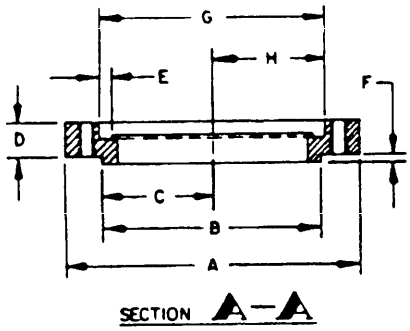


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Preproduction tests(see note 2)		
J	.240 (6.10)	.260 (6.60)
P	.240 (6.10)	.260 (6.60)
Quality conformance inspection, part 1 (see note 4)		
E	8.6825(220.54)	8.6925(220.79)
F	5/16-18UNC-2A	(10) Holes
G	5.4325(137.99)	5.4420(138.23)
K	.245Dia(6.22)	.255Dia(6.48)
L		7.250 (184.15)
M	.240 (6.10)	.260 (6.60)
N		3.475 (88.27)
R	Dr. size S-.348 (8.84) (10) Holes	
T		6.725 (170.82)
U	6.150 (156.21)	6.250 (158.75)
V	.562 (14.27)	
Quality conformance inspection, part 2		
A	4.618(117.30)	4.634(117.70)
B	2.492 (63.30)	2.508 (63.70)
C	2.367 (60.12)	2.383 (60.53)
D	3.930 (99.82)	3.946(100.23)
H		3.3125(84.14)
Q	4.105(104.27)	4.125(104.78)
S	7.355(186.82)	7.375(187.33)

FIGURE 1. Outline drawing of electron tube type T186.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
N	2.309(58.65)	2.317 (58.85)
P	1.246(31.65)	1.254 (31.85)
R	2.371(60.22)	2.379 (60.43)
W	3.934(99.92)	3.942(100.13)
	Nominal	
A	5.437(138.10)	
B	4.000(101.60)	
C	2.000 (50.80)	
D	.600 (15.24)	
E	.2188 (5.56) TYP	
F	.080 (2.03)	
G	4.135(105.03)	
H	2.067 (52.50)	
J	3.537 (89.84)	
K	1.768 (44.91)	
L	.2188RAD(5.56)x.270(6.86)DEEP TYP	
M	8.6875(220.66)	
Q	.3125-18UNC-2B (10) Holes	
S	3.393 (86.18)	
T	6.787(172.39)	
U	3.692 (93.78)	
V	7.385(187.58)	

FIGURE 2. Mounting seat.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1232C ELECTRON TUBE, GAS SWITCHING TYPE 7166

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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